

Fault classification in power system distribution network integrated with distributed generators using CNN

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ABSTRACT

Fault detection is the critical stage of the relaying system and their successful completion in minimum time is expected for fault clearance. With the increasing usage of distributed generators (DGs) in a distribution network, the conventional relaying methods are becoming inappropriate due to changing fault current levels. This paper presents a deep learning algorithm i.e. Convolutional Neural Network (CNN) customized for fault classification in the distributed networks integrated with DGs. This is first time that CNN has been used for fault detection using raw and sampled-data of three-phase voltage and current signals of various fault classes and no-fault class. The 10-fold cross-validation is used to demonstrate the performance of the proposed model in terms of different metrics such as accuracy, sensitivity, specificity, precision, and F1 score. The proposed model has attained an average 10-fold cross-validation accuracy of 99.52% for all the tested fault cases. This featureless proposed method has been compared with conventional approaches from literature and has shown better performance in terms of accuracy and computation burden. Further, a similar fault study is conducted on a mixed transmission line and distribution network with PV as DG using the proposed method and found performance accuracy of 99.92% and 99.97%, respectively.

1. Introduction

In the present time, the demand for electrical power supply is increasing day by day. This has brought in the use of distributed generators (DGs) into the distribution systems to meet out the local demands [1,2]. The integration of these DGs in the distribution network has some benefits like reduction in environmental pollution, efficient power generation, and reduced losses in the transmission lines. Moreover, the voltage profiles get improved and consumers receive a good quality power supply. They have a lower smaller size and capacity of production volume of production and has a lower cost of generation than conventional methods of power generation involving massive generators [3].

On the contrary, they also change the fault current levels and thus disturbs the coordination with protective switchgear [1–3]. The fault current levels vary according to the type of Distributed Energy Resource (DER) connected in the system. The penetration of DGs into distribution networks results in mal-operation of conventional relays in detecting and classifying faults [4,5]. The DER like photovoltaic (PV) is becoming very popular across the globe but its insertion introduces a wide range of power quality issues due to nonlinear power electronics devices and loads [6]. Similarly, the emerging use of electric vehicles penetrating

the bidirectional Electric Vehicle Charging Stations (EVCS) in distribution networks which modifies and harmonizes standard network requirements [7]. Thus, the integration of more and more DERs in the distribution network is creating a challenge to maintain safe and reliable operation of the distribution network [8].

To protect the power system network, it is necessary that once a fault occurs it should be detected and classified accurately as soon as possible so that the faulty phase can be disconnected and there would be a minimum interruption to the network [9]. Here, digital relays come into the picture, which requires an accurate and fast algorithm to detect and classify fault [10]. In recent years, many research works have witnessed the development of various fault detection, classification, and location algorithms in the power system network [11]. Due to the advancement of signal processing techniques, machine learning and deep learning-based techniques can play a vital role in fault detection, classification, and localization in the power system network.

For microgrid protection, time and frequency based differential schemes are proposed in [12], in which the currents from the buses are taken to generate time-frequency contours by S-transform and differential energy is calculated. A threshold is decided to detect a fault based on the calculated differential energy. In microgrid protection, fault detection and classification using Hilbert-Huang Transform (HHT) and

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Support Vector Machine (SVM) based machine learning approach is proposed in [13], in which three-phase current signals are retrieved through buses. Then different intrinsic mode functions (IMFs) are obtained by decomposition and needful features are extracted to provide it to machine learning models to detect and classify faults. Fault classification in a distribution system based on an adaptive neuro-fuzzy inference system (ANFIS) is proposed in [9]. Fast fault detection and classification in distribution lines based on the combination of singular entropy theory and fuzzy logic is proposed in [3]. While features using Teaser energy operators are extracted and two fuzzy inference systems (FIS) are proposed to detect a fault in [1]. In [5], DWT is used to extract features from current signals and then Particle Swarm Optimization is used to find out the optimal wavelet function. This leads to select the most prominent features to be extracted and then conventional machine learning techniques such as k-Nearest Neighbours (KNN), SVM, and Decision Tree (DT) are applied. For fault detection and classification in a microgrid, machine learning and deep learning-based model are proposed in [2,14] and [15] respectively. In [2,14], features are extracted from current signals using wavelet transform, and then it is processed with SVM [2] and DT [14] to classify the fault. Whereas in [15], a combined scheme of Gated Recurrent Unit (GRU) based deep learning model is used for processing current measurements and discrete wavelet transform (DWT) based features as inputs. Fault detection and classification in microgrid using Convolutional Neural Network (CNN) is proposed in [16] where both current and voltage signals are converted into grayscale images and then it is processed with CNN architecture.

In this paper, a deep learning algorithm is proposed for fault classification for the power distribution system integrated with DGs. For the first time, a CNN based deep learning algorithm is proposed for fault classification using raw data samples of three-phase voltage and current signals. The proposed model will be able to classify all types of faults including no-fault as output. The proposed scheme does not require any kind of pre-processing steps as in [15,16]. For example, there is no need for conversion of voltage and current signals into a grayscale image used in [16]. Since the conversion of voltage and current signal into the grayscale image will only increase an avoidable step and use of 2D convolutional layers will also increase the complexity in terms of memory and time. There is also no need for feature-engineering steps, which is a pre-requisite for the conventional machine learning algorithms. The raw samples of voltage and current signals are directly processed as input for the proposed scheme. The relevant and hierarchical level features are automatically extracted by convolutional layers of the proposed CNN model.

The rest of the paper is organized as follows. Section 2 introduces the proposed CNN model used for the classification of the fault. In Section 3, a description of the studied system and the process for data generated from the simulated model is presented. Section 4 is about system implementation and training detail. In Section 5, the performance of the proposed model for fault multi-classification is presented. Further, the discussions on the proposed model are carried out in Section 6. Finally, the conclusions and future scope of work are presented in Section 7.

2. Methodology

In this study, a CNN model is proposed to classify all types of faults including no-fault in a distribution network integrated with DGs. The schematic block diagram of the proposed methodology is illustrated in Fig. 1. The generated data consists of the raw sampled data of three-phase Voltage and Current for different fault and no-fault cases with the corresponding label. In Fig. 1, it can be observed that there is two different phases for the proposed methodology. The top one is the training phase while the bottom one is the testing phase. Initially, the generated datasets are divided into training (i.e. 90% of the dataset) and testing (i.e. 10% of the dataset) cases. During the training phase,

the proposed CNN model is trained using a training dataset along with corresponding actual labels in the offline mode. The proposed CNN model does the feature extraction process automatically from the raw data in this phase. This phase provides the trained weight or trainable parameters of the proposed model. While in the testing phase, the proposed model works in online mode. This phase does not include the feature extraction process and processed the raw data directly as input. The proposed CNN model uses the trained weight or trainable parameters to predict the label of the unseen or new data (i.e. from test case) as output. Lastly, the performance metrics of the proposed CNN model are evaluated by using the predicted label and the actual label of the test case.

The proposed CNN model consists of a total of 9 layers (L_0 - L_8) out of which 4 are weight layers as shown in Table 1. The schematic diagram of the proposed CNN model is depicted in Fig. 2. The input to the proposed model is raw sampled data of three-phase voltage and current (i.e. 6 parameters) readings from one cycle (i.e. 64 samples). Therefore, the input layer L_0 is of shape 64×6 . The raw input data is directly fed to the CNN model without any pre-processing.

In the proposed CNN model, two convolutional stacked layers [17] are used in which each stacked layer has two 1-dimensional (1D) convolutional layers. The main purpose of the convolutional layers is to extract features automatically from the raw input data during offline mode. The Batch Normalization (BN) is used with each convolutional layer to reduce the saturation of the gradient during the covariate shift as it normalizes input features in a batch [18]. The activation function is a non-linear transformation of input. It is a mathematical gate between the input, current neuron, and output. Since the negative values of current and voltage play an important role in the fault classification task. Therefore, the activation function used in each convolutional layer is $\tanh$ function. The range of $\tanh$ is from -1 to 1 . In $\tanh$ function, the negative value of input data is strongly mapped as negative while in ReLU all negative values become zero immediately.

Convolutional layers uses convolution mathematical operation while other neural networks use matrix multiplication operation. Mathematically, the 1D convolution operation is represented by Eq. (1) as follows:

$$C_j = f \left(\sum_{i=1}^N \tilde{X}_{i+j} \times W_i + b \right) \quad (1)$$

where; C is convolution output, $f(\cdot)$ is the $\tanh$ activation function, X and W are mini-batch of input data and filter respectively, is $BN(X)$, N is spatial dimension of filter and b is bias term. The 1D convolutional layers at L_1 , L_2 , L_4 , and L_5 have a set of learnable filters. The number of filters used in the first and second stacked convolutional layer is 16 and 32 respectively. The next layers of convolution with a doubled number of the filter helps in extracting deeper features in hierarchical manners. The convolutional filter of smaller size i.e. 1×3 in dimension is considered. As when there is high dimensional input then it is impractical to connect neurons to all previous volumes. The filter slides over the entire volume of the input and the dot product between the filter and input volume are computed at every spatial position. For avoiding any loss of information, the padding and stride value in the convolutional operation is set such that the output dimension of the convolutional layer remains the same as its input layer. When convolutional layers receive input, then it extracts features automatically based on the weight and biases of the filter and then give output as input to the next layer.

At L_3 and L_6 , 1D Maxpooling (MP) layer [19] is used, which has a kernel size of 1×2 . Maxpooling is a pooling operation that downsamples and highlights the most relevant feature in the receptive field. Before applying max pooling, the network will be looking for a larger area at a time. It downsamples the dimension of convolution output without losing important attributes of the feature maps. It calculates the maximum or largest value from each receptive field. The network picks

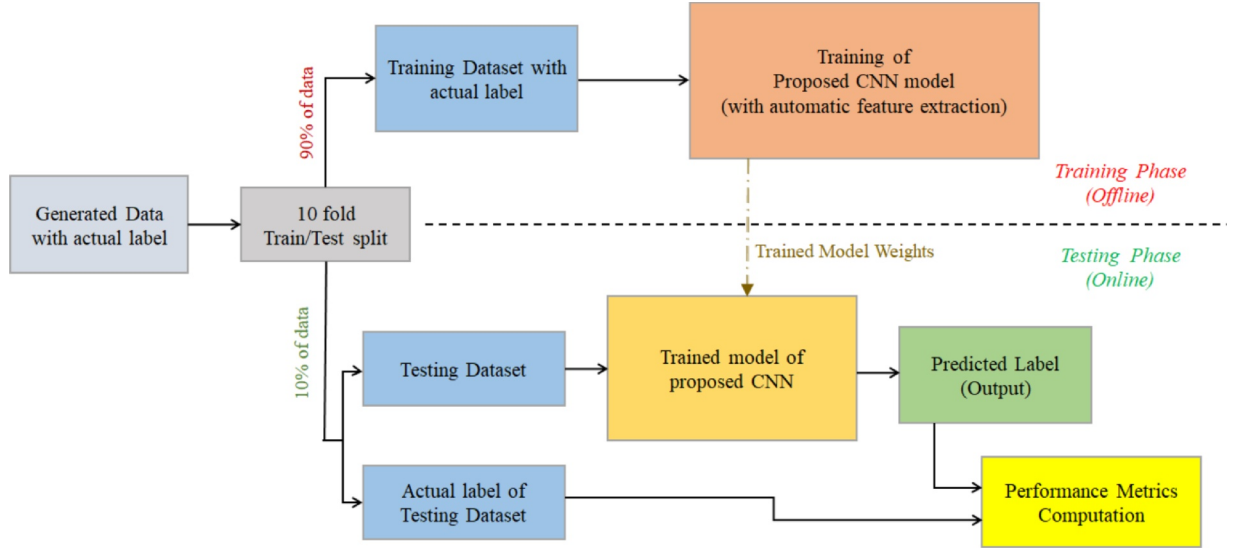


Fig. 1. Block diagram of the proposed methodology.

Table 1
Proposed CNN model architectural details.

Layer no.	Layer detail	Kernel size	Output shape	Trainable parameters
L ₀	Input Raw Data Sample	–	64 × 6	–
L ₁	Conv _{1D} + BN + tanh	1 × 3	64 × 16	304
L ₂	Conv _{1D} + BN + tanh	1 × 3	64 × 16	784
L ₃	MP _{1D}	1 × 2	32 × 16	–
L ₄	Conv _{1D} + BN + tanh	1 × 3	32 × 32	1568
L ₅	Conv _{1D} + BN + tanh	1 × 3	32 × 32	3104
L ₆	MP _{1D}	1 × 2	16 × 32	–
L ₇	Flatten	–	512	–
L ₈	Dense + Softmax	–	11	5643

Conv_{1D} – 1D Convolution layer, BN – Batch normalisation, MP_{1D} – 1D Maxpooling layer.

the maximum value within the selected filter size and sends it to the next layer. The 1D MP operation is defined by Eq. (2) as follows:

$$MP_j = \max(C_j) \quad (2)$$

In this way, filter slides across the whole width and height of the input volume and this reduced the dimension in the output. This also leads to reduce the computational load or burdens. It is also helpful to reduce the overfitting problem.

The CNN process starts with convolution operation, maxpooling, analyzing features, and at last result of these processes are feed to fully connected (FC) layer that drives the final classification decision. The FC layers (L₇-L₈) are used in our proposed model which connects each input neuron to each output neuron within the layer. FC layer takes the output of the previous layer by flattening them, which converts it into a single vector. Therefore, the output from L₆ is flattened at L₇ to a single vectored feature map. The convolution layers before the FC layer have information about the local features in the form of a matrix or higher dimensional map. Whereas, the FC layers have the composite and aggregated information from all the previous layers in a simple vector format. The last layer (L₈) is the output layer in which the number of nodes is equal to the number of classes i.e. 11. A Softmax function is used in the output layer which is used to provide the probability score for each class between 0 and 1. The Softmax function is mathematically represented by Eq. (3) as follows:

$$S(X)_j = \frac{e^{x_j}}{\sum_{k=0}^K e^{x_k}} \quad (3)$$

where; $S(X)_j$ is the probability obtained for j class, x is the output of the previous layer, and e is an exponential function. The class label predicted by the proposed CNN model as output is decided based on the maximum probability score by using Eq. (4) given as follows:

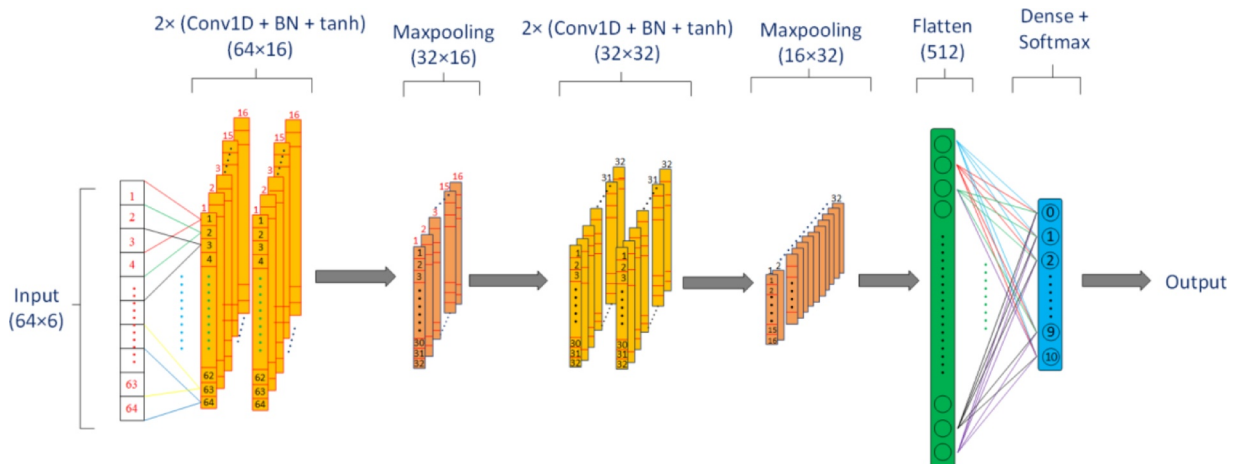


Fig. 2.. Schematic diagram of the proposed CNN model.

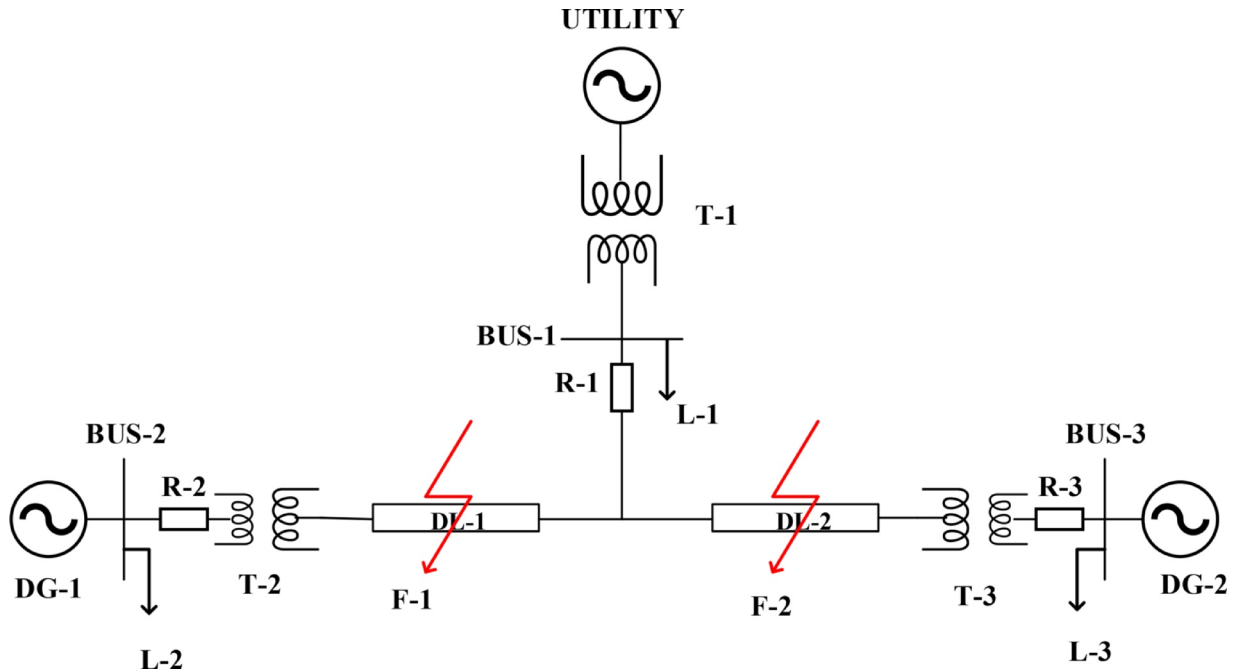


Fig. 3. Single line diagram of distribution network integrated with DG's.

$$P_{class} = \text{argmax}(S) \quad (4)$$

where; P_{class} is predicted class label. Thus, the class having a maximum probability score is the predicted class label.

During training, the network calculates error or loss function using predicted and actual labels. The model tries to minimize the loss function using a weight optimizer. Based on loss function and weight optimizer, the network backpropagates and trainable parameters of the CNN model gets updated to minimize the loss function. Once the model is trained and updated with best model weights or trainable parameters which results in a minimum loss. Then, the proposed model with trained weights processed raw input data directly for fault classification in the online mode without any automatic or manual feature extraction process.

3. Studied system & data generation

In this paper, the distribution network integrated with DGs is used as a studied system. The different parameters of the system are taken as in [1,3]. The single line diagram of the distribution lines integrated with DGs is shown in Fig. 3. The system studied here is modeled in MATLAB/SIMULINK environment. The model is designed for a base frequency of 60 Hz. The studied system contains two DGs, each having 9 MW capacity, are connected to Bus-2 and Bus-3. There are three transformers in the system where the transformer T-1 steps down the voltage from 66KV to 20KV (66KV/20KV) while transformer T-2 and T-3 steps up the voltage from 400 V to 20KV (400 V/20KV). There are two pi-section lines DL-1 and DL-2 having a length of 25 Km each. The rating of the utility grid is 1000MVA working on 66 KV voltage. There are three loads in the system Load-1 (L-1), Load-2 (L-2), and Load-3 (L-3) connected to Bus-1, Bus-2, and Bus-3 respectively. L-1 is a reactive load of 6 MW and + 2 MVAR rating while L-2 and L-3 are resistive loads of 3 MW each. If there is any fault in the distribution lines (either in DL-1 or DL-2) then Relay-1 will sense it and Bus-1 or PCC will disconnect it by isolating the Utility grid and distribution line. Moreover, the tripping of Bus-2 and Bus-3 will lead to operating the DG's to supply their local loads only.

The three-phase voltage and current signal data are generated from the simulation model of the studied system. The three-phase voltage and current measurements are sampled by Relay R-1 that can able to

detect a fault in both distribution lines DL-1 & DL-2. The sampling frequency for relay R-1 is taken as 3.84 kHz [15]. Therefore, the total no. of samples in one cycle will be $(3840 \div 60) = 64$ samples. The different types of faults and no-fault cases are simulated to generate a large number of datasets to train and check the versatility of the proposed model. This is achieved by setting different configurations and parameters of the modeled system. The different possible configuration for fault and non-fault cases is tabulated in Tables 2 and 3 respectively. The total number of cases in each class with their corresponding label is outlined in Table 4. The number of cases in each of the fault class is 2688 (i.e. $2 \times 24 \times 7 \times 8$) and the total non-fault generated cases are 2197 (i.e. $13 \times 13 \times 13$). Therefore, the total number of sampled data generated for 11 class are 29,077 (i.e. $(2688 \times 10) + 2197$).

The different types of fault are simulated with all possible configuration of fault resistances and fault inception angle at different lines and location of the studied system. The fault resistance with low or high-value settings shows different characteristics. By varying the fault resistances, there would be a different fault current level. The fault can occur at any angle from 0° to 180° with different characteristics. The signal waveforms of three-phase voltage and current generated for different simulated fault cases with different fault inception angle are shown in Fig. 4. The fault is created at $t = 3$ s with different fault inception angle. The distorted waveform of voltage and the current signal after a fault at different inception angle can easily be noted in Fig. 4. If a fault occurs at 0° , then all 64 samples in one cycle of data will

Table 2
Configuration for fault cases simulation.

Parameter	Possible configuration	Count
Fault type	AG, BG, CG, AB, BC, AC, ABG, BCG, ACG, ABC	10
Fault line	DL-1, DL-2 (each having distance of 25 Km)	2
Fault location(in Km)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	24
Fault Inception angle(in degree)	0, 30, 60, 90, 120, 150, 180	7
Fault resistance(in Ω)	0.01, 0.2, 2, 5, 10, 20, 50, 100	8

Table 3
Configuration for non-fault cases simulation.

Parameter	Possible configuration	Count
Load-1(L1)	$\pm (0, 10, 20, 30, 40, 50, 60)\%$ of change in load	13
Load-2(L2)	$\pm (0, 10, 20, 30, 40, 50, 60)\%$ of change in load	13
Load-3(L3)	$\pm (0, 10, 20, 30, 40, 50, 60)\%$ of change in load	13

Table 4
Total cases in each class.

Class Label	Class Name	No. of cases
0	No-Fault (NF)	2197
1	AG	2688
2	BG	2688
3	CG	2688
4	AB	2688
5	BC	2688
6	AC	2688
7	ABG	2688
8	BCG	2688
9	ACG	2688
10	ABC	2688

be fault data. If a fault occurs at 45° , then $(1/8)$ th of total samples i.e. 8 samples are from non-fault data and 56 samples from fault data. While if a fault occurs at 180° then 32 samples from non-fault and 32 samples from fault data. All the sampled data of one cycle (i.e. 64 samples) after the occurrence of fault irrespective of fault inception angle is processed as input for the proposed scheme.

4. System implementation and training

The proposed model has been implemented using Python and Keras library [20] with TensorFlow [21] backend engine. The time series simulated data is generated using MATLAB/SIMULINK. All the experiments are performed on a system having an Intel CORE i5 8250 U processor with 8GB RAM.

The 10-fold cross-validation method instead of a custom train/test split is used. Thus, the entire cases have been subdivided into 10 uniform subsets and further 9 subsets (i.e. 90% of the cases) are used for training and remaining 1 subset (i.e. 10% of the cases) for testing. At the outset, the weights are initialized using Xavier's method [22]. Adam optimizer [23] is used to train the model with an initial learning rate of 0.001. The adaptive learning rate is used in which the learning rate is gradually reduced by a factor of 0.1 until learning stagnates. The categorical cross-entropy is used as loss functions. The total number of epochs used for training the model is 50 with a batch size of 10.

5. Results

The performance of the proposed model has been evaluated using various metrics such as accuracy, recall, specificity, precision, and F1-score. The formulae of these performance metrics are tabulated below in Table 5.

As mentioned earlier, the 10-fold cross-validation has been used to validate the proposed model. The 10-fold cross-validation result of the proposed model is presented in Table 6. The performance metrics obtained in each fold are almost similar. The average value of each performance metrics for all folds are also calculated and mentioned in Table 6. From Table 6, it can be observed that the proposed model provides superior performance for fault classification in the studied system.

The value of different performance metrics of Fold No. 2 (i.e. F2) is less compared to other folds. The class-wise performance metrics of the proposed model for F2 are given in Table 7. From Table 7, it can be

observed that the accuracy in each class is above 0.99. However, the F1-score of No-fault (NF), BC, and BCG faults is about 0.97 as the number of misclassification of test cases is more compared to other classes.

The accuracy and loss curves for both training and testing cases over 50 epochs for Fold No. 2 (i.e. F2) are shown in Figs. 5 and 6 respectively. It can be observed from Figs. 5 and 6 that with the initial learning rate, the testing curve tries to match the training curve. But at the epoch around 20, a plateau in model performance can be seen. This indicates that the initial learning rate will take more epoch for convergence of the model with higher performance. As the model is suffering a lot to learn the problem with the initial learning rate. Therefore, the adaptive learning strategy reduces the learning rate at that point for achieving good model performance on the train and test sets with rapid convergence. After the change in learning rate (at around epoch 20), the sudden improvement can be seen in the model performance. The learning rates gradually reduces further with defined factor until learning stagnates. Finally, the proposed model converges with higher performance after 30 epochs as the training curve overlaps with the testing curve. It is a sign that there is no need for further reduction of learning rate and more training epochs.

6. Discussions

6.1. Performance behavior of CNN for fault classification

For studying the performance behavior of the CNN model for fault classification, the different CNN configurations are used with a different selection of the number of convolutional layers. The comparison between the performances of different CNN models with the proposed model is tabulated in Table 8. All the experiments have been implemented with the same train/test split and in the same computing platform environment. From Table 8, it can be observed that the performance for fault classification increases as the convolutional layers increases. But the more number of convolutional layer leads to an increase in the number of trainable parameters. The use of stacked convolutional layers used in the proposed model shows superior performance along with less number of trainable parameter. Thus, the proposed model maintains the trade-off between model performance and complexity in terms of trainable parameters.

6.2. Comparison of the proposed method with other methods

The comparative analysis of the proposed scheme with different existing methods is also performed with the data generated from the studied system. The comparison is done based on both classification performance and computational cost. The proposed model is compared with different machine learning methods such as SVM [2,5], DT [5,14], and KNN [5] classification model. The wavelet-based features [2,5,14,15] are extracted from three-phase current and voltage measurement samples which are used as input for considered machine learning models. The proposed model is also compared with the GRU model [15], which is another state-of-the-art deep learning model. The raw input and voltage samples are processed as input to the GRU model similarly as in the proposed model. The performance and computational cost of considered existing methods are compared with the proposed model and tabulated in Table 9. All the experiments have been implemented with the same train/test split ratio (i.e. 9:1) and in the same computing platform environment. The training and testing data consists of 26,169 and 2908 cases.

From Table 9, it can be observed that the accuracy of the proposed model is comparable when compared to other models. But, the accuracy of the DT [5,14] is higher than the proposed model. However, the same cannot be concluded for the computational time. During the training phase, machine learning models take less computational time when compared to GRU and the proposed CNN model. It shows that deep learning algorithms are computationally expensive during the training

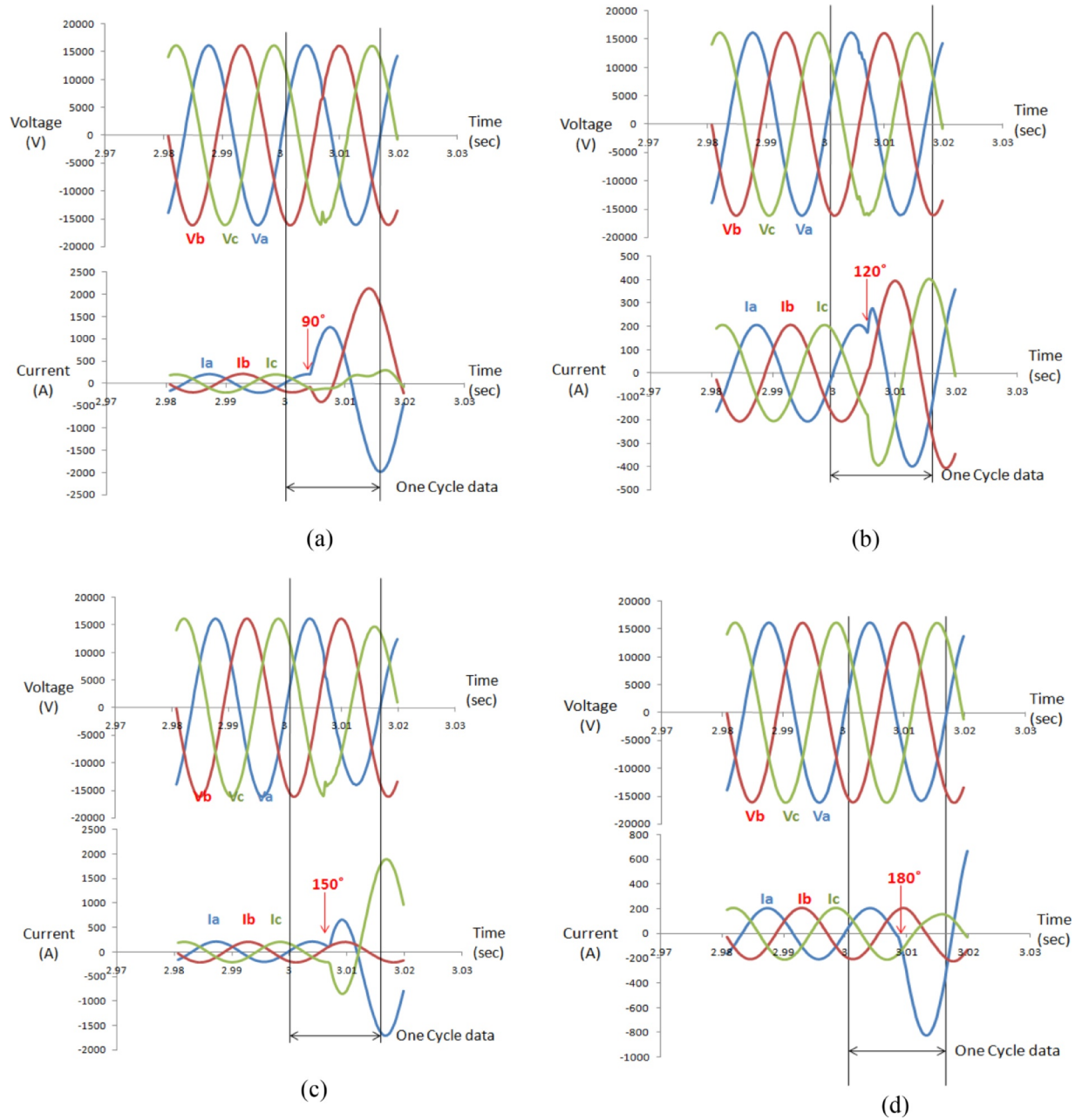


Fig. 4. Voltage and current signal waveforms with different inception angle. (a) ABG fault, (b) ABC fault, (c) AC fault, and (d) AG fault.

Table 5
Different performance metrics formulae.

Sl. no.	Performance Metric	Formulae
1	Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$
2	Recall	$\frac{TP}{TP + FN}$
3	Specificity	$\frac{TN}{TN + FP}$
4	Precision	$\frac{TP}{TP + FP}$
5	F1 – score	$2 \times \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$

TP– True Positive, TN – True Negative, FP– False Positive, FN – False Negative.

of the model. As the deep learning-based GRU and proposed CNN models extract features automatically during training. However, the training of the models is an offline process. The models are implemented for the intended task in online mode (i.e. test phase) only

Table 6
10-fold cross-validation result of the proposed model for fault classification.

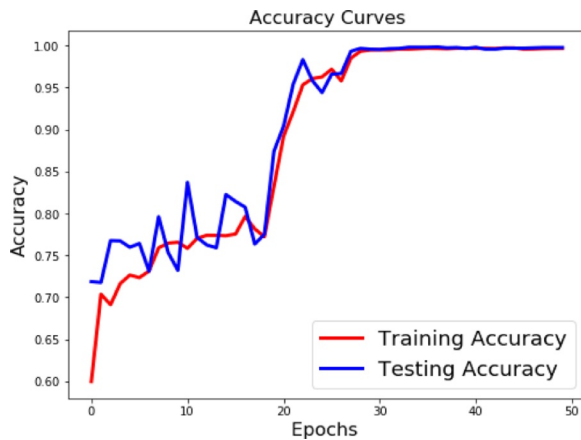
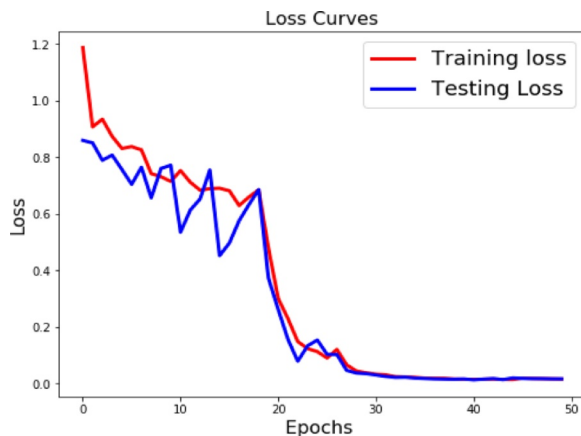
Fold No.	Accuracy	Recall	Specificity	Precision	F1-Score
F0	0.9966	0.9966	0.9996	0.9966	0.9966
F1	0.9976	0.9976	0.9997	0.9976	0.9976
F2	0.9890	0.9890	0.9989	0.9893	0.9890
F3	0.9976	0.9976	0.9991	0.9976	0.9976
F4	0.9973	0.9973	0.9997	0.9973	0.9973
F5	0.9900	0.9900	0.9992	0.9904	0.9901
F6	0.9983	0.9983	0.9998	0.9983	0.9983
F7	0.9962	0.9962	0.9996	0.9963	0.9962
F8	0.9948	0.9948	0.9994	0.9949	0.9948
F9	0.9945	0.9945	0.9995	0.9946	0.9945
Average	0.9952	0.9952	0.9995	0.9953	0.9952

Table 7

Class-wise result of the proposed model for fold no. 2.

Class Name	Accuracy	Recall	Specificity	Precision	F1-Score	FN	FP
NF	0.9948	0.9545	0.9981	0.9767	0.9655	10	5
AG	1.0000	1.0000	1.0000	1.0000	1.0000	0	0
BG	0.9996	0.9963	1.0000	1.0000	0.9981	1	0
CG	1.0000	1.0000	1.0000	1.0000	1.0000	0	0
AB	0.9983	0.9851	0.9996	0.9962	0.9907	4	1
BC	0.9945	0.9963	0.9943	0.9470	0.9710	1	15
AC	0.9979	0.9963	0.9981	0.9817	0.9889	1	5
ABG	1.0000	1.0000	1.0000	1.0000	1.0000	0	0
BCG	0.9945	0.9442	0.9996	0.9961	0.9695	15	1
ACG	1.0000	1.0000	1.0000	1.0000	1.0000	0	0
ABC	0.9983	1.0000	0.9981	0.9817	0.9908	0	5
Wt. Avg.	0.9890	0.9890	0.9989	0.9893	0.9890	–	–

Wt. Avg. – Weighted average.

**Fig. 5.** Accuracy curve of fold no. 2.**Fig. 6.** Loss curve of fold no. 2.

after the training of the model to predict the fault class label of 2908 unseen or test cases. Once these deep learning models are trained, then there is no need for features or feature extraction in the online mode.

Table 8

Results of different CNN models for fault classification in the studied system.

CNN Models	Trainable parameters	Accuracy	Sensitivity	Precision	F1-Score
2 × (Conv + MP) + FC + Dense	7515	0.9773	0.9773	0.9782	0.9775
3 × (Conv + MP) + FC + Dense	13,723	0.9800	0.9800	0.9800	0.9800
4 × (Conv + MP) + FC + Dense	38,427	0.9840	0.9840	0.9840	0.9840
Proposed model	11,403	0.9988	0.9988	0.9988	0.9988

Conv – Conv1D + BN + tanh.

Table 9

Comparison of different methods w.r.t accuracy and computational time.

Classification model	Accuracy	Training phase (Offline)		Testing phase (Online)	
		Feature Extraction (in sec)	Training time(in sec)	Feature Extraction (in sec)	Testing time(in sec)
SVM [2,5]	0.9961	406.87	124.23	48.21	4.70
KNN [5]	0.9902	406.87	0.02	48.21	0.097
DT [5,14]	0.9990	406.87	0.15	48.21	0.006
GRU [15]	0.9979	Nil	1350.42	Nil	4.57
Proposed model (CNN)	0.9988	Nil	882.16	Nil	1.21

They directly processed the raw sample data as input with trained weights of the proposed model to predict the fault case label. Whereas, the machine learning models need extra computational time and effort for manual feature extraction. As they process features as input with trained parameters which is extracted from raw sample data. For example; DT with the highest accuracy takes 48.216 s (i.e. 48.21 + 0.006) of total computational time during the testing phase. DT model takes 48.21 s just for extracting features to process it as input. Whereas, the proposed model with comparable performance takes only 1.21 s of total computational time during the testing phase. Therefore, the total computational time during the testing phase (in online mode) for the proposed model is much less than the conventional machine learning models. Hence, the proposed CNN model is computationally efficient in the online mode when compared to conventional machine learning models. From Table 9, it can also be concluded that the proposed CNN model outperforms the GRU model [15] in terms of both performance and computational cost.

6.3. Performance on other system and with different DGs

The proposed CNN model is also applied on a mixed transmission line and a distribution line having PV as DG to test its efficacy on other systems for fault multi-classification. The system details and fault classification performance with the proposed algorithm of both of these considered systems are given below.

- Mixed Transmission Line:** The mixed line parameters are considered from [24]. Fig. 7 represents the single line diagram of a mixed transmission line. The mixed transmission line is 15 Km out of which 3 Km is underground cable and 12 Km is an overhead line. Relay-1 is placed near to Bus-1 and can sense fault from both lines (i.e. underground cable and overhead line). The total of 10,240 fault and non-fault cases are generated out of which 90% are taken as train cases and 10% are taken as test cases. The performance of the proposed model on the mixed transmission line is tabulated in Table 10.
- Power System Distribution Line with PV as DG:** The distribution line system have 400 KW PV farm as DG which is connected to Bus-2 through a transformer of 260 V/25KV. Fig. 8 represents the single line diagram of this considered system. There are three buses in the system and between these buses, two distribution lines are

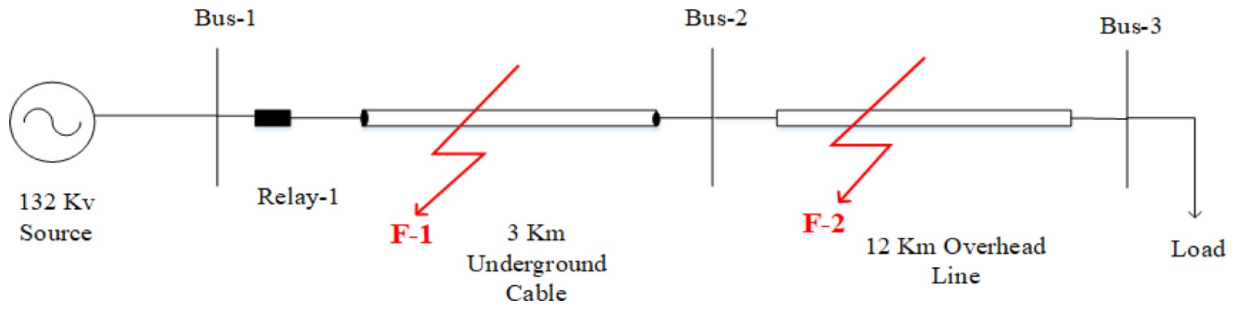


Fig. 7. Single line diagram of Mixed Transmission Line.

Table 10

Performance of the proposed model on Mixed Transmission line.

Accuracy	Recall	Specificity	Precision	F1-Score
0.9992	0.9992	0.9998	0.9992	0.9992

Table 11

Performance of the proposed model on distribution line having PV as DG.

Accuracy	Recall	Specificity	Precision	F1-Score
0.9997	0.9997	0.9999	0.9997	0.9997

connected having a length of 14 Km each. There are three loads on each of the respective buses. L-1 is of 200 KW, +20 KVAR rating connected to Bus-1. L-2 is of 100 KW, −40 KVAR rating connected to Bus-2. And load-3 is of 200 KW rating connected to Bus-3. The sampling frequency of the system is chosen as 3.84 kHz and the system is working on a 50 Hz system. Considering Relay-1 for sensing the fault in the system, a total of 9100 fault cases and 1331 non-fault cases are generated. Out of 10,431, 90% is taken as train cases and 10% are taken as test cases. The performance of the proposed model on the distribution line with PV as DG is tabulated in Table 11.

From Tables 10 and 11, it can be concluded that the proposed algorithm achieves an accuracy of more than 99% in both of the considered systems. The different type of DGs does not affect the performance of the proposed algorithm. Thus, the proposed algorithm for fault classification can apply to other systems also with a different type of DGs too.

7. Conclusion

In this paper, fault classification in the power system distribution network integrated with DGs using CNN has been proposed. The proposed model does not require any pre-processing or feature engineering steps, which makes them reliable and computationally efficient during the online or testing period. The performance of the proposed model was extensively evaluated using 10-fold cross-validation, which makes the proposed model more generalized and robust for fault classification.

The outstanding performance for fault classification in the studied system is achieved by the proposed model like 10-fold cross-validation accuracy of 99.52%. The proposed deep learning architecture is compared with conventional machine learning methods and the GRU model in terms of both performance and computational cost. The proposed model shows comparable performance and also found to be computationally efficient during online mode w.r.t. other machine learning models. However, the training of the proposed model in the offline mode is computationally expensive. The proposed CNN model with less trainable parameters is found to be better in performance and computational cost when compared to other existing deep learning model i.e. GRU. The proposed model is also tested on a mixed line and distribution line with PV as DG and shows efficient performance. Thus, the proposed method can be used for fault classification in other power system networks also. In the future, the proposed deep learning-based methodology can be discovered for fault localization tasks. As a scope for further research, the shallow CNN model with efficient performance and computation cost will be designed and implemented for fault identification in the real distribution grid system.

CRediT authorship contribution statement

Praveen Rai: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **Narendra D. Londhe:** Supervision, Project administration, Validation, Resources. **Ritesh Raj:** Software, Writing - original draft.

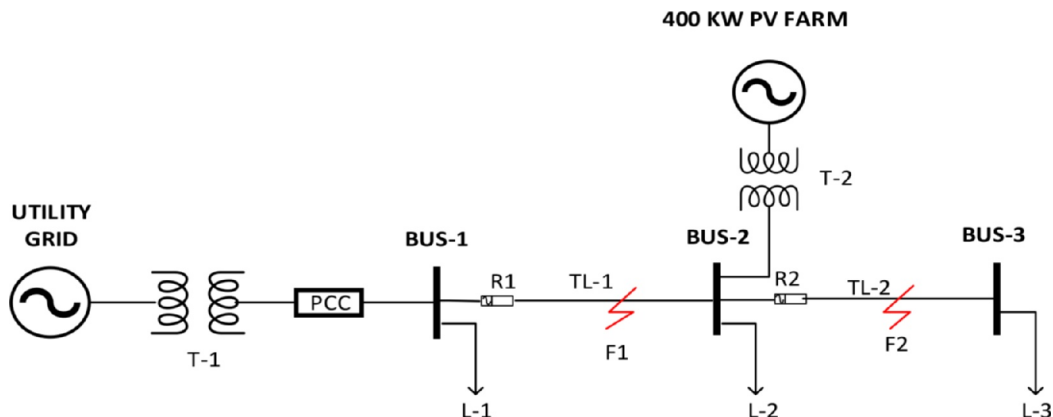


Fig. 8. Single line diagram of the power system distribution line.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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